

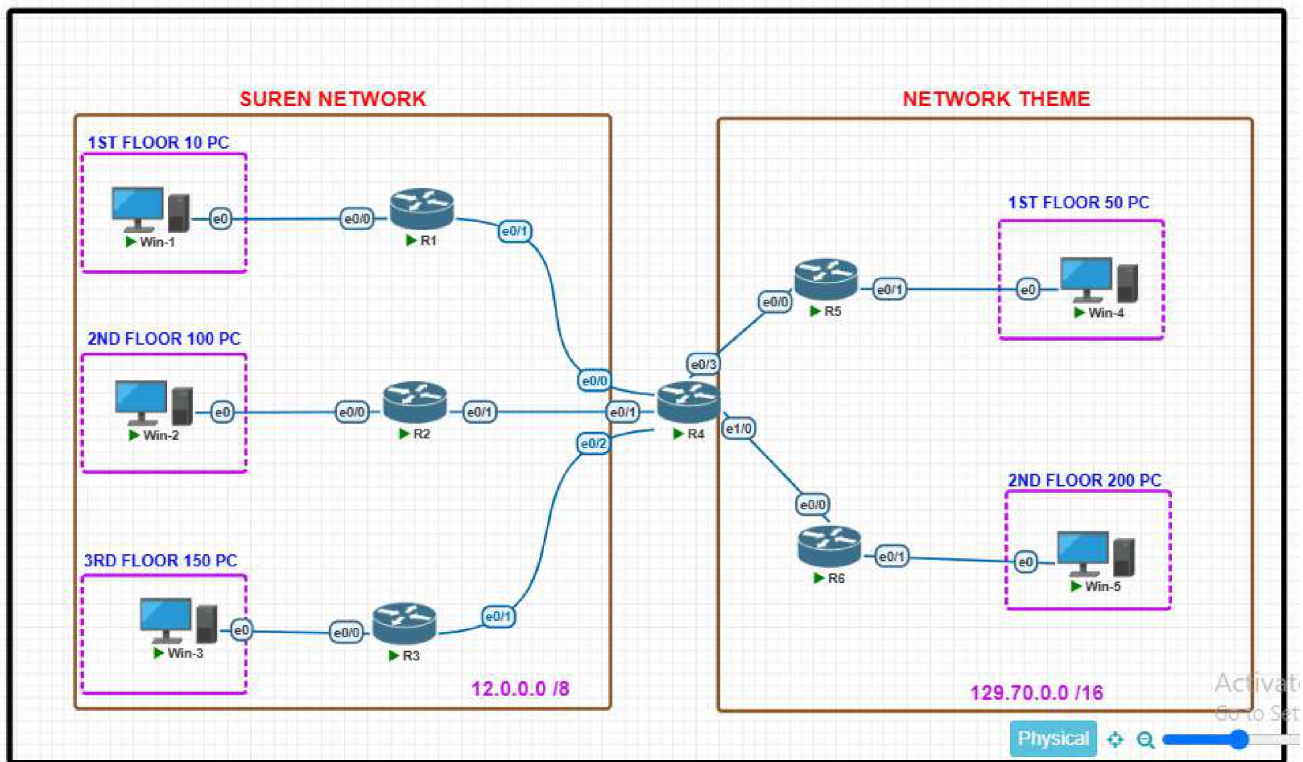
CCNA Practice Project

IPv4 Subnetting, DHCP Services, Routing & Connectivity Testing

Simulation / Lab Evidence Report

Simulation Tool	Cisco Packet Tracer
Core Topics	IPv4 Subnetting • DHCP • Default Gateway • Routing • Connectivity Testing
Project Evidence	19 original screenshots preserved in high resolution

Figure 1 — Original network topology evidence.



1. Project Objective

This project demonstrates a practical IPv4 networking lab in which a network is divided into multiple logical segments. The objective is to apply subnetting to create appropriate IP networks, configure DHCP-based client addressing, verify router interfaces and routes, and prove end-to-end reachability using standard network commands.

Project goals

- Develop an IPv4 addressing plan using subnetting.
- Configure host addressing, subnet masks and default gateways correctly.
- Configure and verify DHCP operation for client devices.
- Configure routing between separate IP networks.
- Validate the completed design with ipconfig, router show commands and ping.

2. IP Addressing & Subnetting

The project uses separate IPv4 networks for different parts of the topology. The evidence shows addressing in the 12.0.0.0/8 and 129.70.0.0/16 ranges, with smaller subnets assigned to individual LAN segments and point-to-point links.

Network / Segment	Observed addressing / purpose
12.0.x.x networks	LAN addressing used by several end devices and routed segments
129.70.x.x networks	Additional LAN/WAN addressing visible in the project evidence
Router-to-router links	Point-to-point addressing used to interconnect routed segments

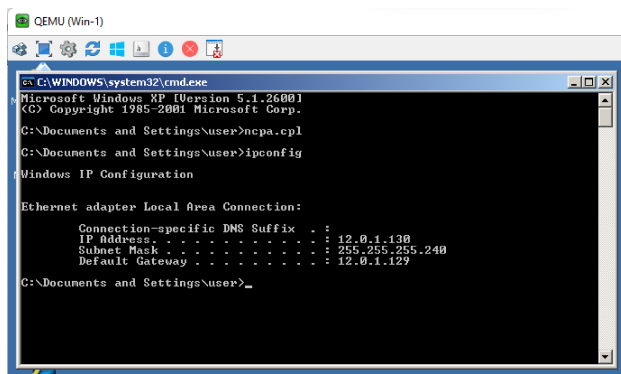
The exact IP values shown in the screenshots are retained as project evidence rather than replaced with generic examples.

3. End-Device IP Configuration (ipconfig)

The Windows command-prompt screenshots show the IP configuration assigned to end devices. These checks are important because a host must have an address, the correct subnet mask and the correct default gateway before it can communicate beyond its local subnet.

Verification performed

- Checked the assigned IPv4 address on each host.
- Confirmed the subnet mask matches the intended network.
- Confirmed the default gateway points to the appropriate router interface.
- Used the output as the first step before testing remote connectivity.



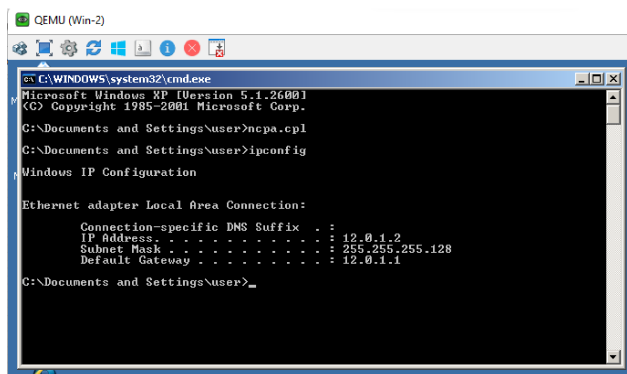
```
QEMU (Win-1)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ipconfig
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 12.0.1.130
    Subnet Mask . . . . . : 255.255.255.240
    Default Gateway . . . . . : 12.0.1.129

C:\Documents and Settings\user>
```

Figure 2 — End-device IP configuration evidence.



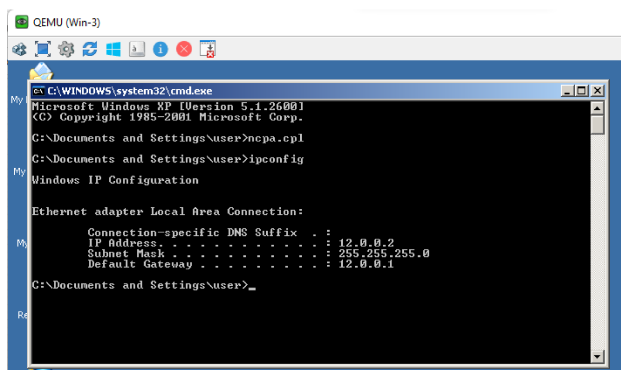
```
QEMU (Win-2)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ipconfig
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 12.0.1.2
    Subnet Mask . . . . . : 255.255.255.128
    Default Gateway . . . . . : 12.0.1.1

C:\Documents and Settings\user>
```

Figure 3 — End-device IP configuration evidence.



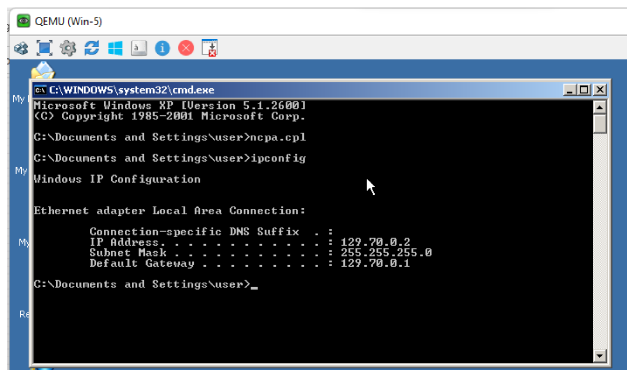
```
QEMU (Win-3)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ipconfig
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 12.0.0.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 12.0.0.1

C:\Documents and Settings\user>
```

Figure 4 — End-device IP configuration evidence.



```
QEMU (Win-5)
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.
C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ipconfig
Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 129.70.0.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 129.70.0.1

C:\Documents and Settings\user>
```

Figure 5 — End-device IP configuration evidence.

3.1 Additional End-Device Verification

The remaining host screenshots provide additional confirmation that addressing was applied consistently across the lab. Together, the screenshots form a practical record of the client-side configuration before and during connectivity testing.

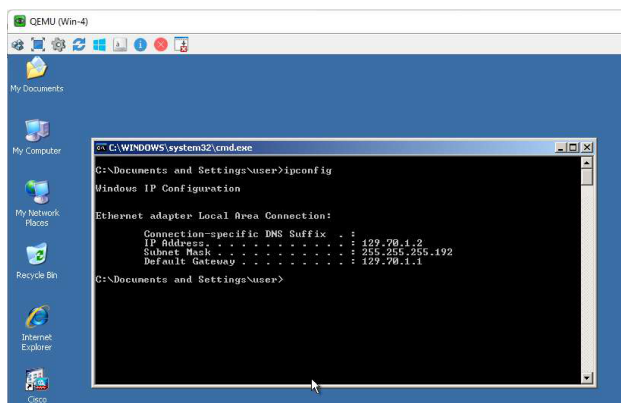


Figure 6 — Additional host-side IP configuration evidence.

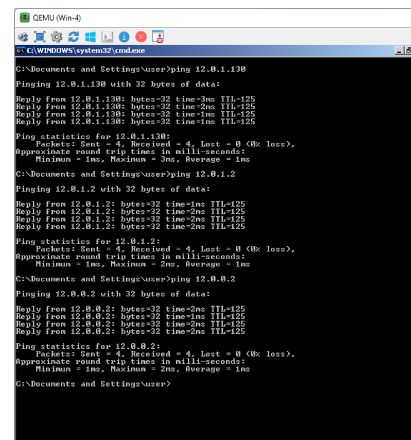


Figure 7 — Additional host/network command evidence.

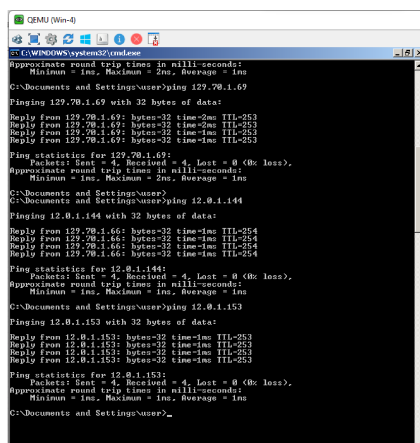


Figure 8 — Additional host/network command evidence.

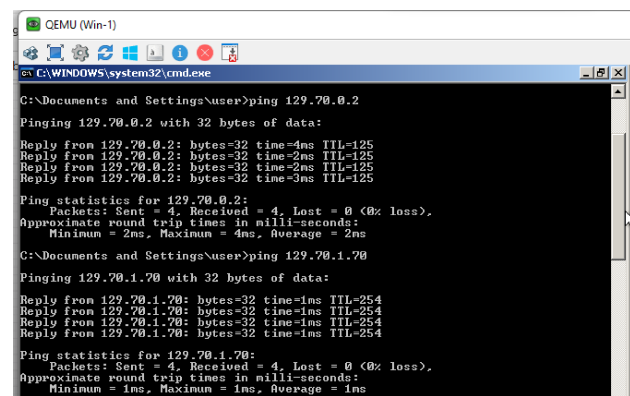


Figure 9 — Additional connectivity evidence.

Professional interpretation

These checks demonstrate a basic but essential troubleshooting workflow: verify Layer-3 host configuration first, then investigate interfaces, routing and reachability if communication fails.

4. Router Configuration Verification

Cisco IOS verification commands are used to confirm that router interfaces are configured and operational and that the routing table contains the networks required for communication. The project evidence includes command-line outputs from the routing infrastructure.

Key verification areas

- Interface addressing and operational status.
- Connected and remote routes.
- DHCP pool and lease information.
- Successful forwarding between separate subnets.

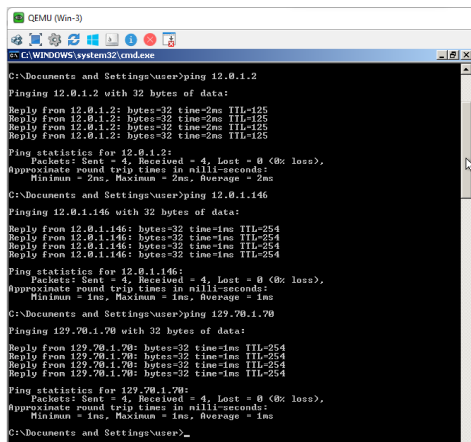


Figure 10 — Router CLI verification evidence.

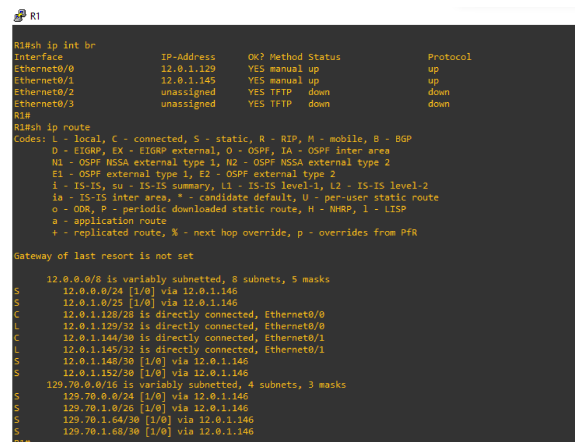


Figure 11 — Router CLI verification evidence.

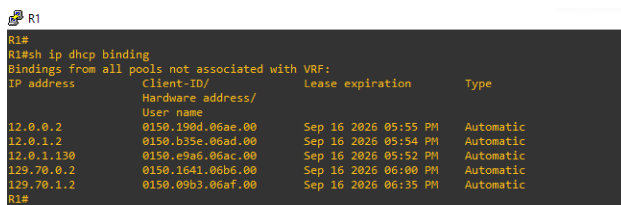


Figure 12 — Router CLI verification evidence.

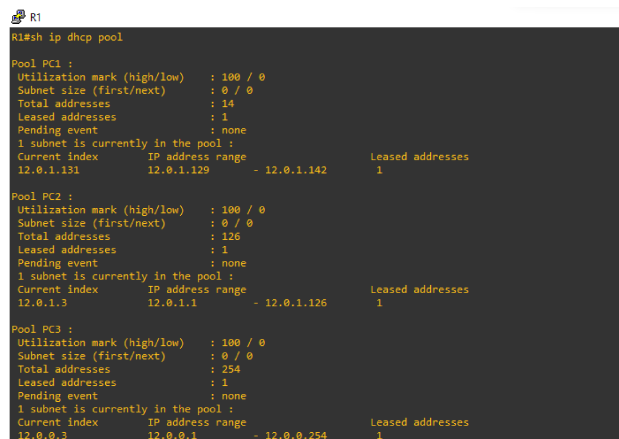


Figure 13 — Router CLI verification evidence.

4.1 DHCP Services & Verification

DHCP is used to automate IPv4 configuration for client devices. The router-side evidence includes DHCP-related output that can be used to confirm pools and active bindings.

DHCP verification approach

- Confirm that the DHCP pool exists for the intended client network.
- Check that leases are being created for client devices.
- Compare the assigned addresses with the expected subnet.
- Confirm clients can use the assigned gateway to reach remote networks.

```
R1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
12.0.1.131 12.0.1.126 - 12.0.1.142 1

Pool PC1 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 126
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
12.0.1.3 12.0.1.1 - 12.0.1.126 1

Pool PC3 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 154
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
12.0.0.3 12.0.0.1 - 12.0.0.254 1

Pool PC4 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 154
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
129.70.1.3 129.70.0.1 - 129.70.0.254 1

Pool PC5 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next) : 0 / 0
Total addresses : 154
Leased addresses : 1
Pending event : none
1 subnet is currently in the pool :
Current index IP address range Leased addresses
129.70.0.3 129.70.0.1 - 129.70.0.254 1
```

Figure 14 — DHCP configuration / verification evidence.

```
R2
R2(config-if)#exit
R2(config)#exit
R2#
*Sep 15 13:14:47.656: NBY-5-COMP10.1: Configured from console by console
R2#sh ip int br
Interface IP-Address OK? Method Status Prot
Ethernet0/0 12.0.1.1 YES manual up
Ethernet0/1 12.0.1.149 YES manual up
Ethernet0/2 unassigned YES TFTP down
Ethernet0/3 unassigned YES TFTP down

R2#
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       Ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - OMR, P - periodic downloaded static route, H - NHRP, I - IGRP
       * - replicated route, N - next hop override, p - overrides from PFH

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
S 12.0.0.0/24 [1/0] via 12.0.1.150
C 12.0.1.0/25 is directly connected, Ethernet0/0
L 12.0.1.1/32 is directly connected, Ethernet0/0
S 12.0.1.128/28 [1/0] via 12.0.1.150
L 12.0.1.144/30 [1/0] via 12.0.1.150
C 12.0.1.148/30 is directly connected, Ethernet0/1
L 12.0.1.149/32 is directly connected, Ethernet0/1
S 12.0.1.152/32 [1/0] via 12.0.1.150
S 129.70.0.0/16 is variably subnetted, 4 subnets, 3 masks
S 129.70.0.0/24 [1/0] via 12.0.1.150
S 129.70.1.0/28 [1/0] via 12.0.1.150
S 129.70.1.64/30 [1/0] via 12.0.1.150
S 129.70.1.68/30 [1/0] via 12.0.1.150
```

Figure 15 — DHCP configuration / verification evidence.

```
R3
Router#enable
Router#sh ip int br
Interface IP-Address OK? Method Status Prot
Ethernet0/0 12.0.0.1 YES manual up
Ethernet0/1 12.0.1.152 YES manual up
Ethernet0/2 unassigned YES unset administratively down
Ethernet0/3 unassigned YES unset administratively down

Router#
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       Ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - OMR, P - periodic downloaded static route, H - NHRP, I - IGRP
       * - replicated route, N - next hop override, p - overrides from PFH

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
C 12.0.0.0/24 is directly connected, Ethernet0/0
S 12.0.0.0/25 [1/0] via 12.0.1.150
L 12.0.1.128/28 [1/0] via 12.0.1.154
L 12.0.1.144/30 [1/0] via 12.0.1.154
S 12.0.1.148/30 [1/0] via 12.0.1.154
L 12.0.1.152/32 is directly connected, Ethernet0/1
L 12.0.1.153/32 is directly connected, Ethernet0/2
S 129.70.0.0/16 is variably subnetted, 4 subnets, 3 masks
S 129.70.0.0/24 [1/0] via 12.0.1.154
S 129.70.1.0/28 [1/0] via 12.0.1.154
S 129.70.1.64/30 [1/0] via 12.0.1.154
S 129.70.1.68/30 [1/0] via 12.0.1.154
```

Figure 16 — DHCP configuration / verification evidence.

```
R4
Router#enable
Router#sh ip int br
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 12.0.1.146 YES manual up
Ethernet0/1 12.0.1.150 YES manual up
Ethernet0/2 12.0.1.154 YES manual up
Ethernet0/3 129.70.1.65 YES manual up
Ethernet0/4 unassigned YES unset administratively down
Ethernet0/5 unassigned YES unset administratively down
Ethernet0/6 unassigned YES unset administratively down

Router#
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       Ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - OMR, P - periodic downloaded static route, H - NHRP, I - IGRP
       * - replicated route, N - next hop override, p - overrides from PFH

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 9 subnets, 5 masks
S 12.0.0.0/24 [1/0] via 12.0.1.155
S 12.0.1.0/25 [1/0] via 12.0.1.149
C 12.0.1.128/28 [1/0] via 12.0.1.146
L 12.0.1.144/30 is directly connected, Ethernet0/0
L 12.0.1.146/32 is directly connected, Ethernet0/0
L 12.0.1.148/30 is directly connected, Ethernet0/1
L 12.0.1.150/32 is directly connected, Ethernet0/1
L 12.0.1.152/32 is directly connected, Ethernet0/2
L 12.0.1.154/32 is directly connected, Ethernet0/3
S 129.70.0.0/16 is variably subnetted, 8 subnets, 4 masks
S 129.70.0.0/24 [1/0] via 129.70.1.65
S 129.70.1.0/28 [1/0] via 129.70.1.65
L 129.70.1.64/30 is directly connected, Ethernet0/3
C 129.70.1.68/30 is directly connected, Ethernet0/0
L 129.70.1.70/32 is directly connected, Ethernet0/4
```

Figure 17 — DHCP configuration / verification evidence.

4.2 Routing Verification

Routing is the mechanism that allows hosts in different IP networks to communicate. The router command outputs in this project provide evidence of interface addressing and route information.

What was verified

- Router interfaces have valid IP addresses.
- Relevant interfaces are operational.
- Local networks appear in the routing table.
- Remote networks are reachable through the configured routing path.

```
Router#show ip int br
+-----+
| Interface | IP-Address | OK? Method Status | Protocol |
+-----+
| Ethernet0/0 | 129.70.1.65 | YES manual up | up |
| Ethernet0/1 | 129.70.1.1 | YES manual up | up |
| Ethernet0/2 | unassigned | YES unset administratively down | down |
| Ethernet0/3 | unassigned | YES unset administratively down | down |
+-----+
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       O - OSPF, EX - OSPF external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, I - IGRP
       * - application route
       + - replicated route, X - next hop override, p - overrides from PFR

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S 12.0.0.0/24 [1/0] via 129.70.1.66
S 12.0.1.0/25 [1/0] via 129.70.1.66
S 12.0.1.128/26 [1/0] via 129.70.1.66
S 12.0.1.144/30 [1/0] via 129.70.1.66
S 12.0.1.148/26 [1/0] via 129.70.1.66
S 12.0.1.152/30 [1/0] via 129.70.1.66
129.70.0.0/16 is variably subnetted, 6 subnets, 4 masks
S 129.70.0.0/24 [1/0] via 129.70.1.66
S 129.70.1.0/26 is directly connected, Ethernet0/1
C 129.70.1.1/32 is directly connected, Ethernet0/1
C 129.70.1.65/32 is directly connected, Ethernet0/0
C 129.70.1.66/30 [1/0] via 129.70.1.66
Router#
```

Figure 18 — Routing/interface verification evidence.

```
Router#show ip int br
+-----+
| Interface | IP-Address | OK? Method Status | Protocol |
+-----+
| Ethernet0/0 | 129.70.1.69 | YES manual up | up |
| Ethernet0/1 | 129.70.0.1 | YES manual up | up |
| Ethernet0/2 | unassigned | YES TFTP down | down |
| Ethernet0/3 | unassigned | YES TFTP down | down |
+-----+
Router#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       O - OSPF, EX - OSPF external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       I - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, I - IGRP
       * - application route
       + - replicated route, X - next hop override, p - overrides from PFR

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S 12.0.0.0/24 [1/0] via 129.70.1.70
S 12.0.1.0/25 [1/0] via 129.70.1.70
S 12.0.1.128/26 [1/0] via 129.70.1.70
S 12.0.1.148/26 [1/0] via 129.70.1.70
S 12.0.1.152/30 [1/0] via 129.70.1.70
S 12.0.1.156/30 [1/0] via 129.70.1.70
129.70.0.0/16 is variably subnetted, 6 subnets, 4 masks
C 129.70.0.0/24 is directly connected, Ethernet0/1
L 129.70.0.0/25 is directly connected, Ethernet0/1
C 129.70.1.0/26 [1/0] via 129.70.1.70
C 129.70.1.68/30 [1/0] via 129.70.1.70
C 129.70.1.69/32 is directly connected, Ethernet0/0
Router#
```

Figure 19 — Routing/interface verification evidence.

The screenshots are reproduced without modification so the original CLI evidence remains available for technical review.

5. End-to-End Connectivity Testing

After addressing, DHCP and routing were configured, connectivity was tested using ICMP ping. A successful ping confirms that the destination is reachable and that the intervening Layer-3 path is functioning.

Testing method

1. Confirm the source host has a valid IPv4 address and gateway.
2. Ping a local gateway to confirm basic local connectivity.
3. Ping a host on another subnet to test routing.
4. Repeat from multiple hosts to confirm bidirectional reachability.
5. Use packet-loss results to identify whether further troubleshooting is required.

Evidence sequence

The original project contains multiple command-prompt and router CLI screenshots. They are all preserved in the evidence section of this report.

6. Technical Evidence — Screenshot 1

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.

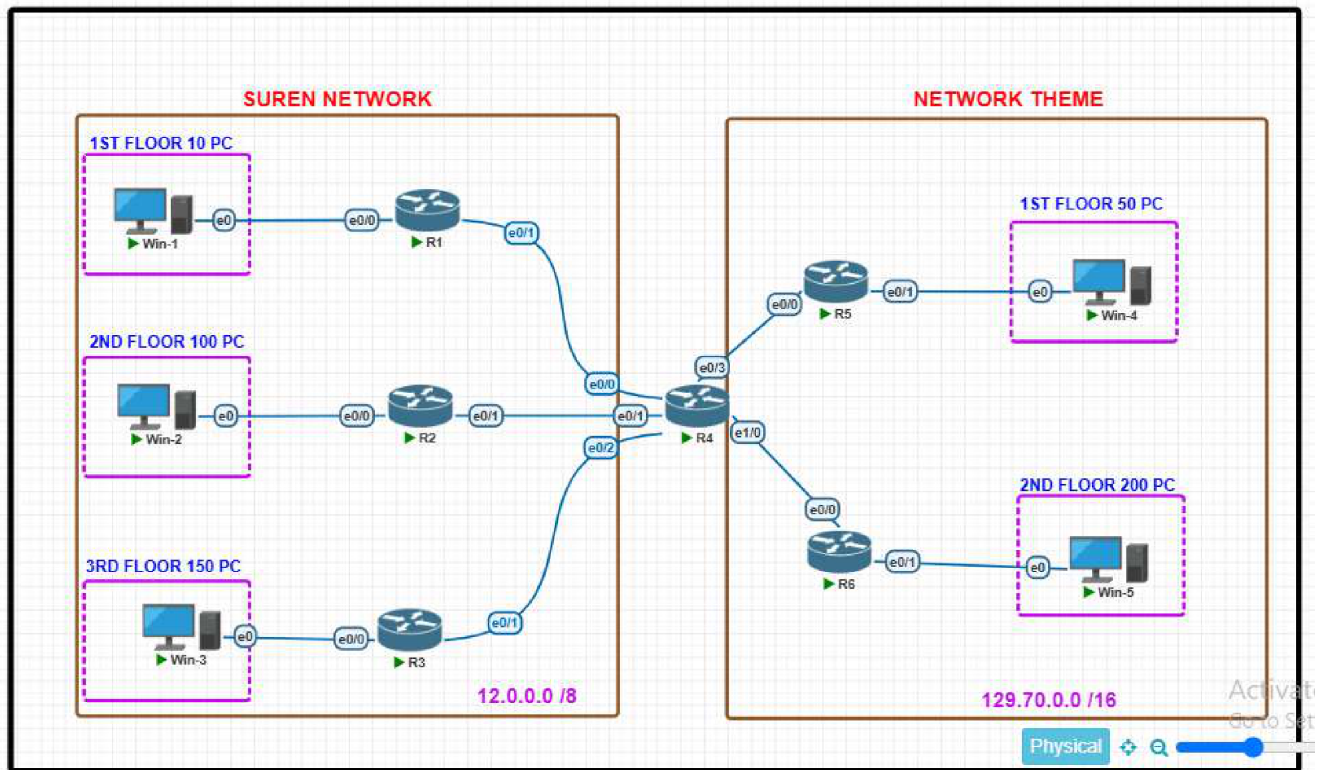
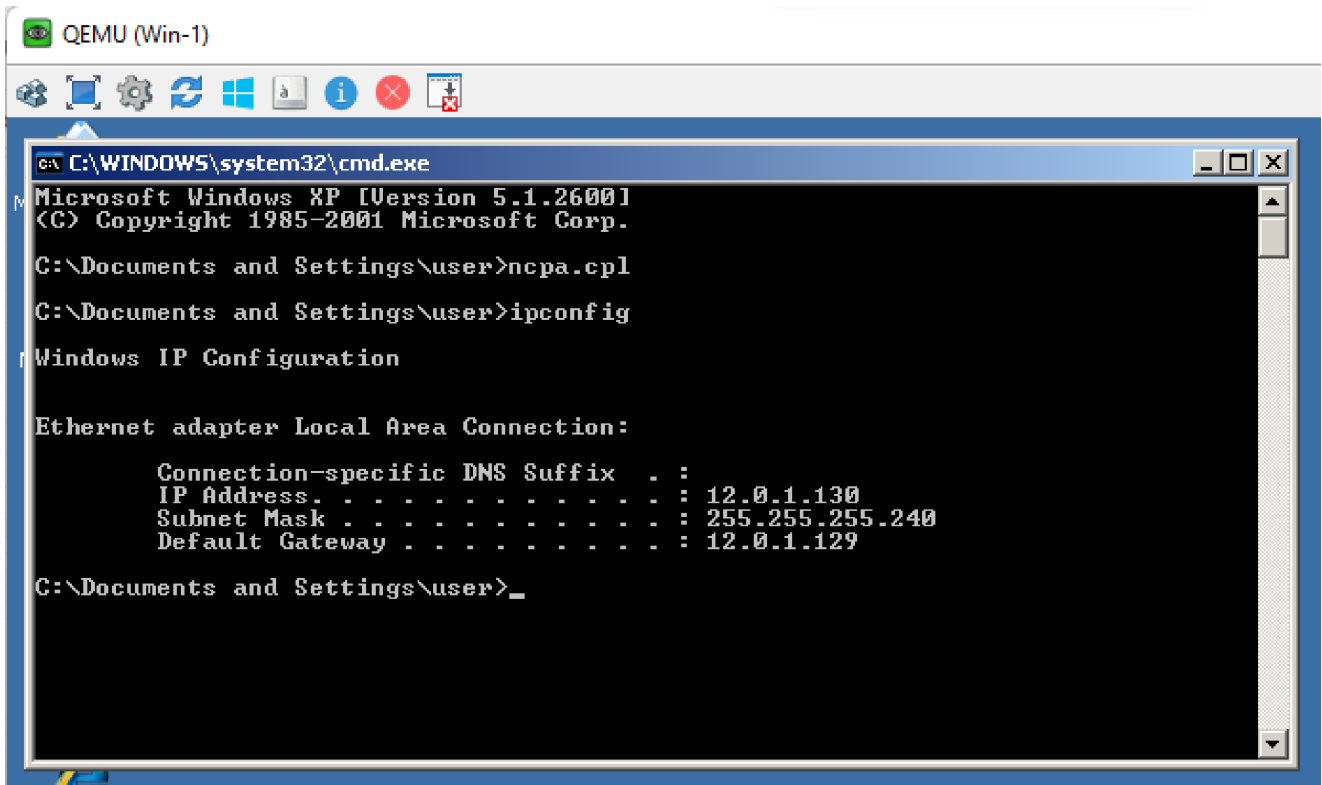


Figure 1 — Original project screenshot.

6. Technical Evidence — Screenshot 2

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

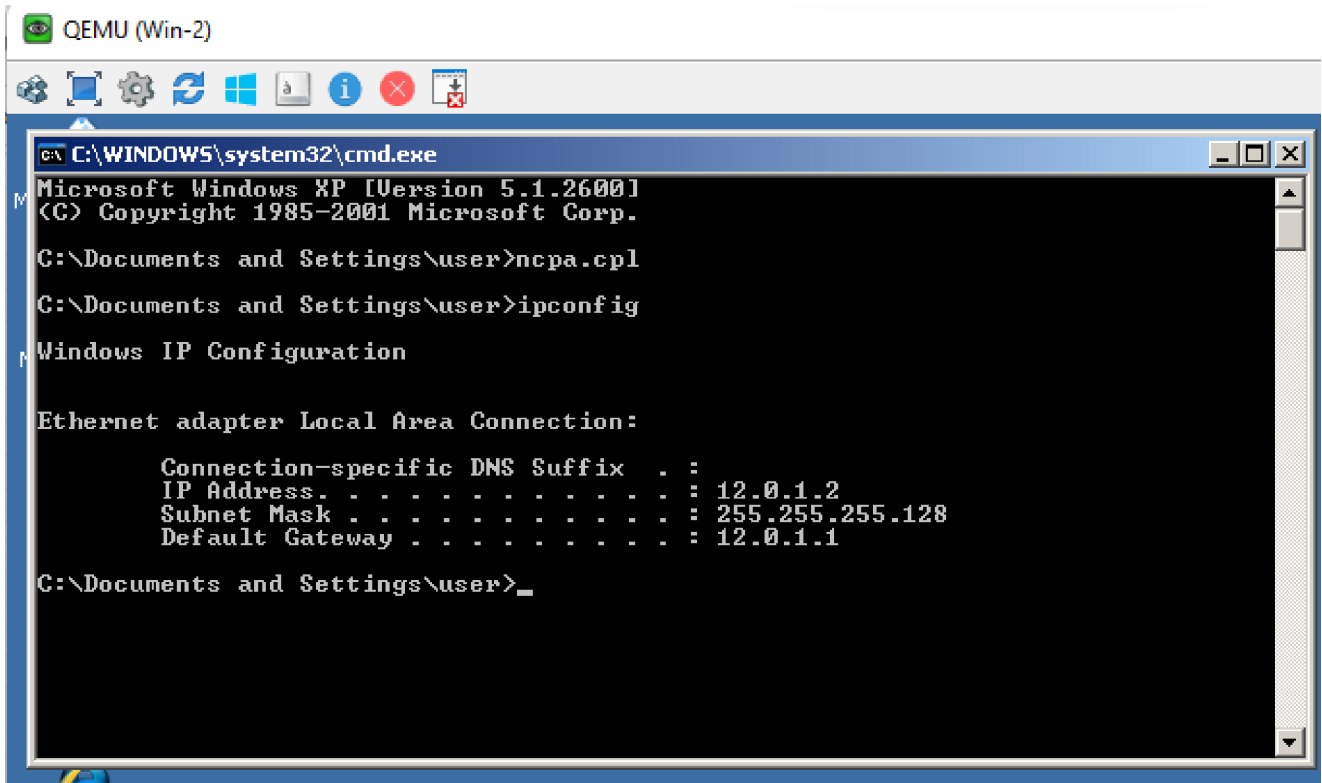
    Connection-specific DNS Suffix  . : 
    IP Address. . . . .               : 12.0.1.130
    Subnet Mask . . . . .             : 255.255.255.240
    Default Gateway . . . . .         : 12.0.1.129

C:\Documents and Settings\user>
```

Figure 2 — Original project screenshot.

6. Technical Evidence — Screenshot 3

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



```
C:\WINDOWS\system32\cmd.exe
Microsoft Windows XP [Version 5.1.2600]
(C) Copyright 1985-2001 Microsoft Corp.

C:\Documents and Settings\user>ncpa.cpl
C:\Documents and Settings\user>ipconfig

Windows IP Configuration

Ethernet adapter Local Area Connection:

    Connection-specific DNS Suffix  . : 
    IP Address. . . . . : 12.0.1.2
    Subnet Mask . . . . . : 255.255.255.128
    Default Gateway . . . . . : 12.0.1.1

C:\Documents and Settings\user>
```

Figure 3 — Original project screenshot.

6. Technical Evidence — Screenshot 4

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.

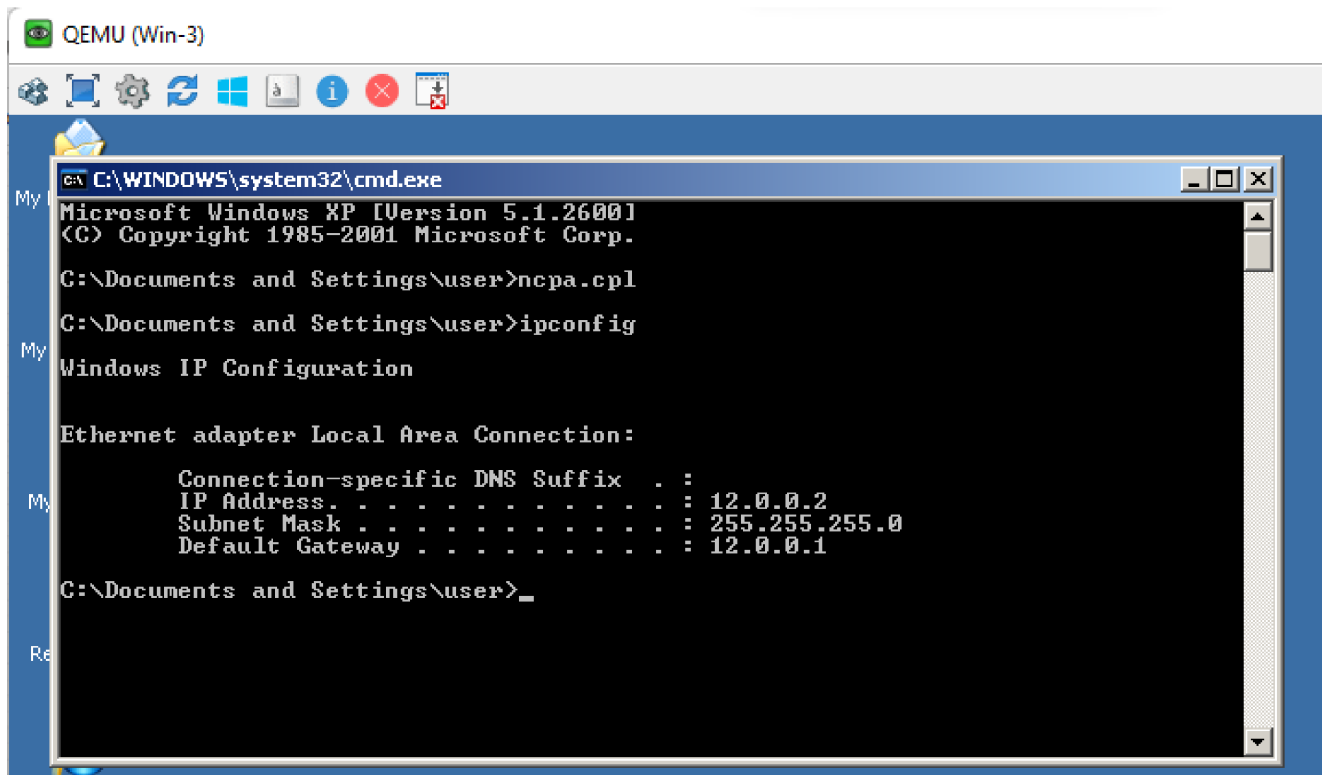


Figure 4 — Original project screenshot.

6. Technical Evidence — Screenshot 5

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.

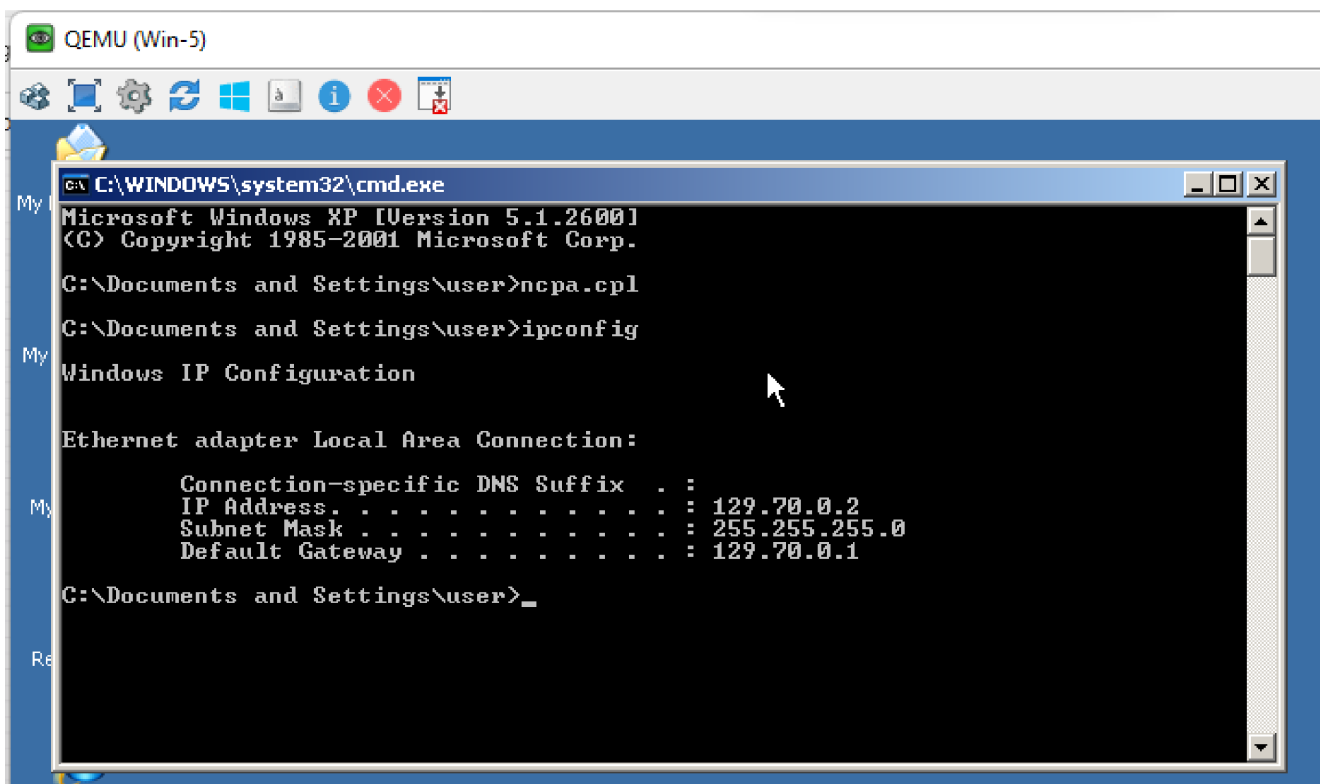


Figure 5 — Original project screenshot.

6. Technical Evidence — Screenshot 6

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.

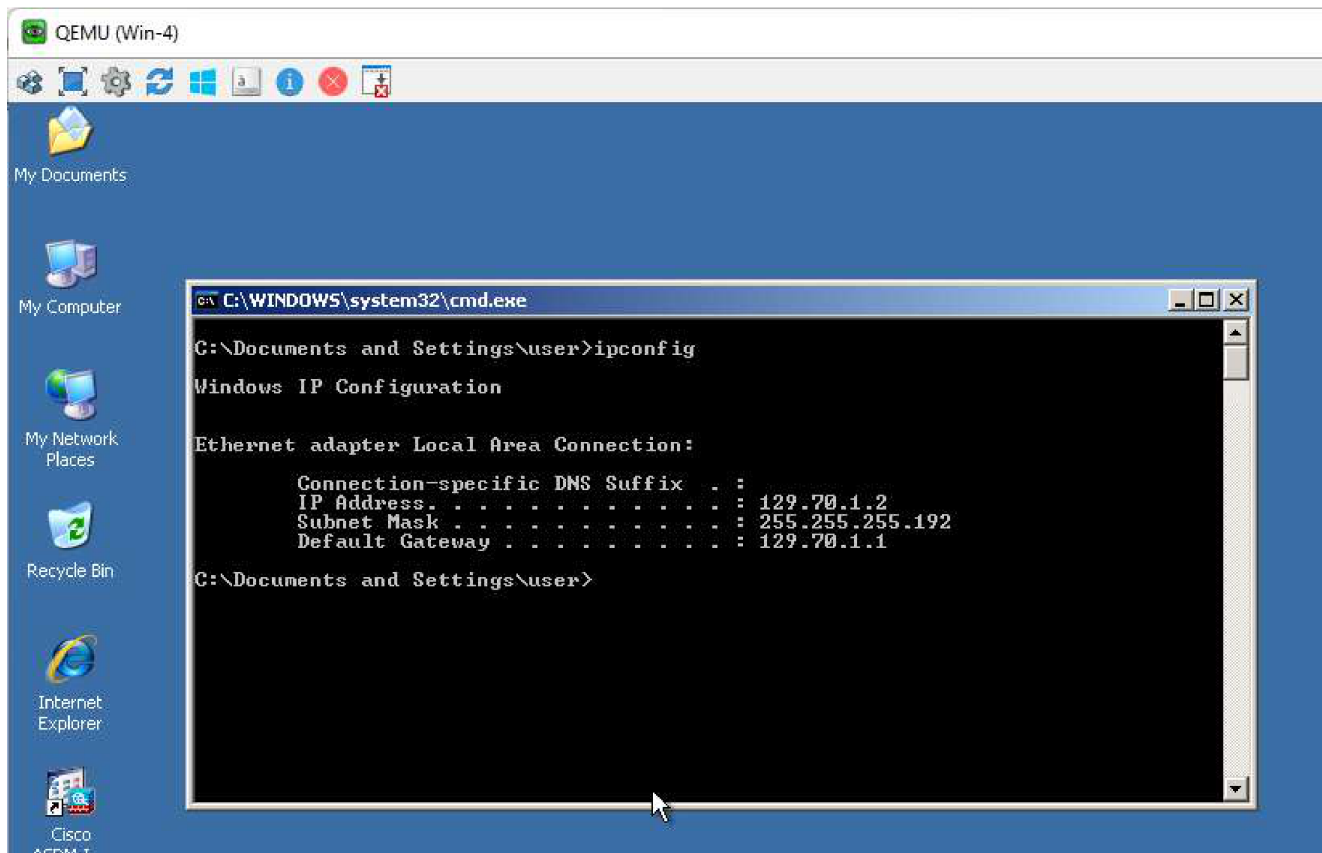
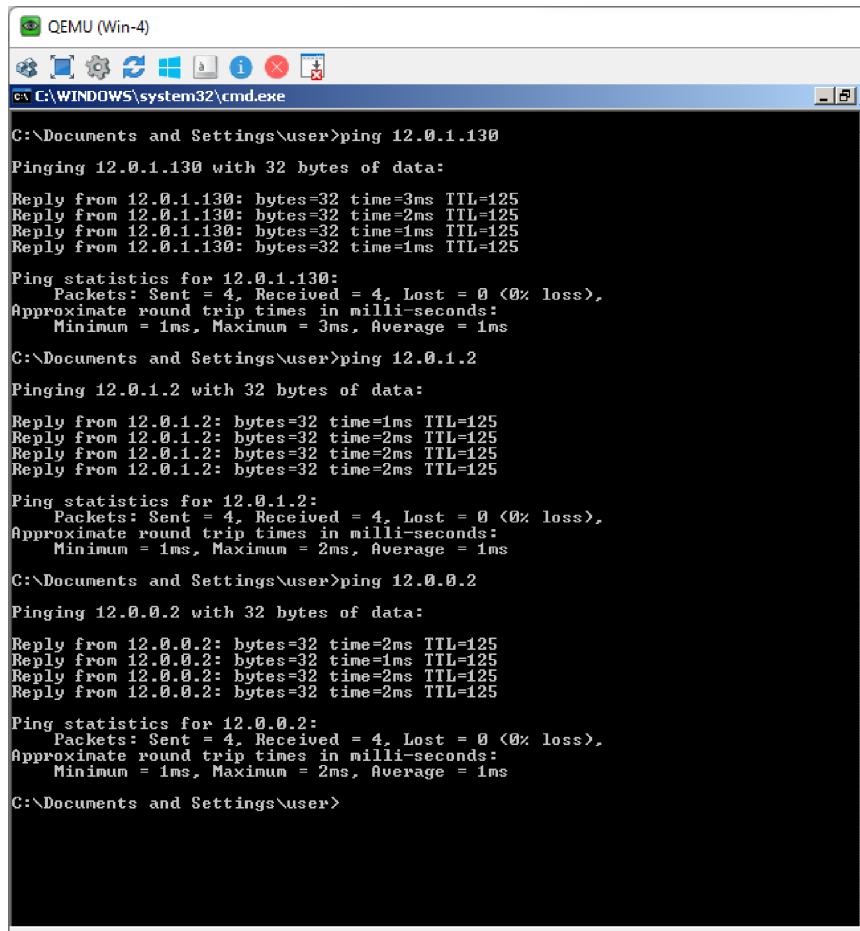


Figure 6 — Original project screenshot.

6. Technical Evidence — Screenshot 7

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.

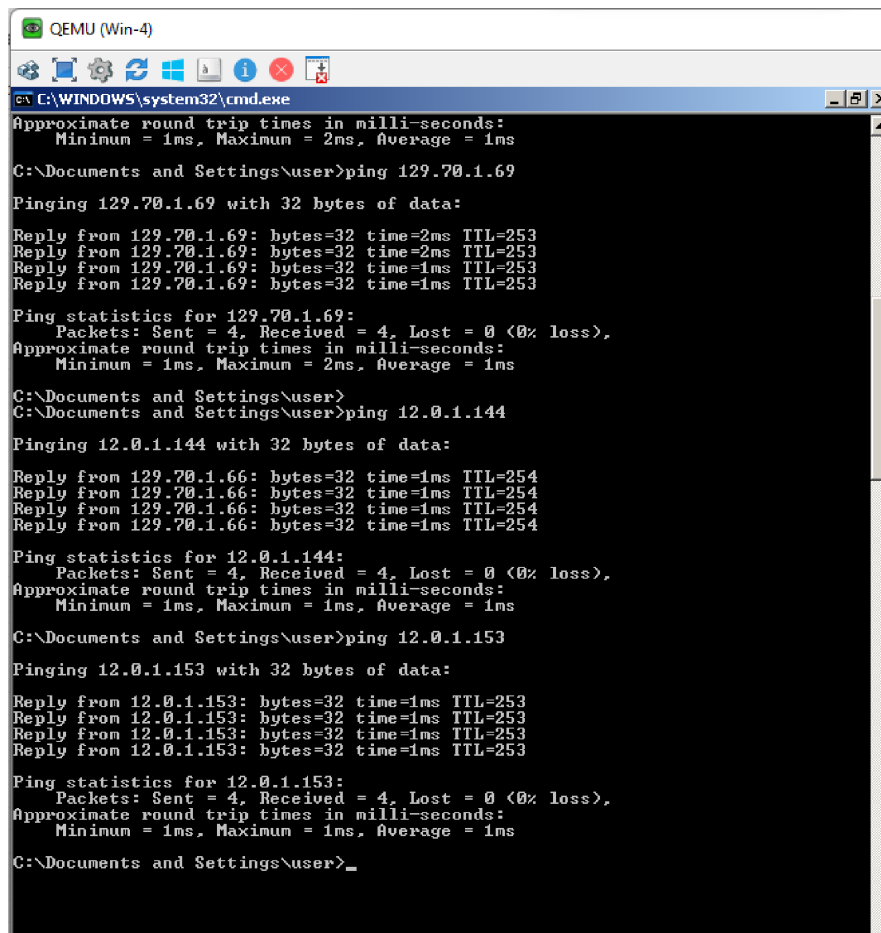


```
QEMU (Win-4)
C:\WINDOWS\system32\cmd.exe
C:\Documents and Settings\user>ping 12.0.1.130
Pinging 12.0.1.130 with 32 bytes of data:
Reply from 12.0.1.130: bytes=32 time=3ms TTL=125
Reply from 12.0.1.130: bytes=32 time=2ms TTL=125
Reply from 12.0.1.130: bytes=32 time=1ms TTL=125
Reply from 12.0.1.130: bytes=32 time=1ms TTL=125
Ping statistics for 12.0.1.130:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms
C:\Documents and Settings\user>ping 12.0.1.2
Pinging 12.0.1.2 with 32 bytes of data:
Reply from 12.0.1.2: bytes=32 time=1ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Ping statistics for 12.0.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>ping 12.0.0.2
Pinging 12.0.0.2 with 32 bytes of data:
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=1ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Reply from 12.0.0.2: bytes=32 time=2ms TTL=125
Ping statistics for 12.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>
```

Figure 7 — Original project screenshot.

6. Technical Evidence — Screenshot 8

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.

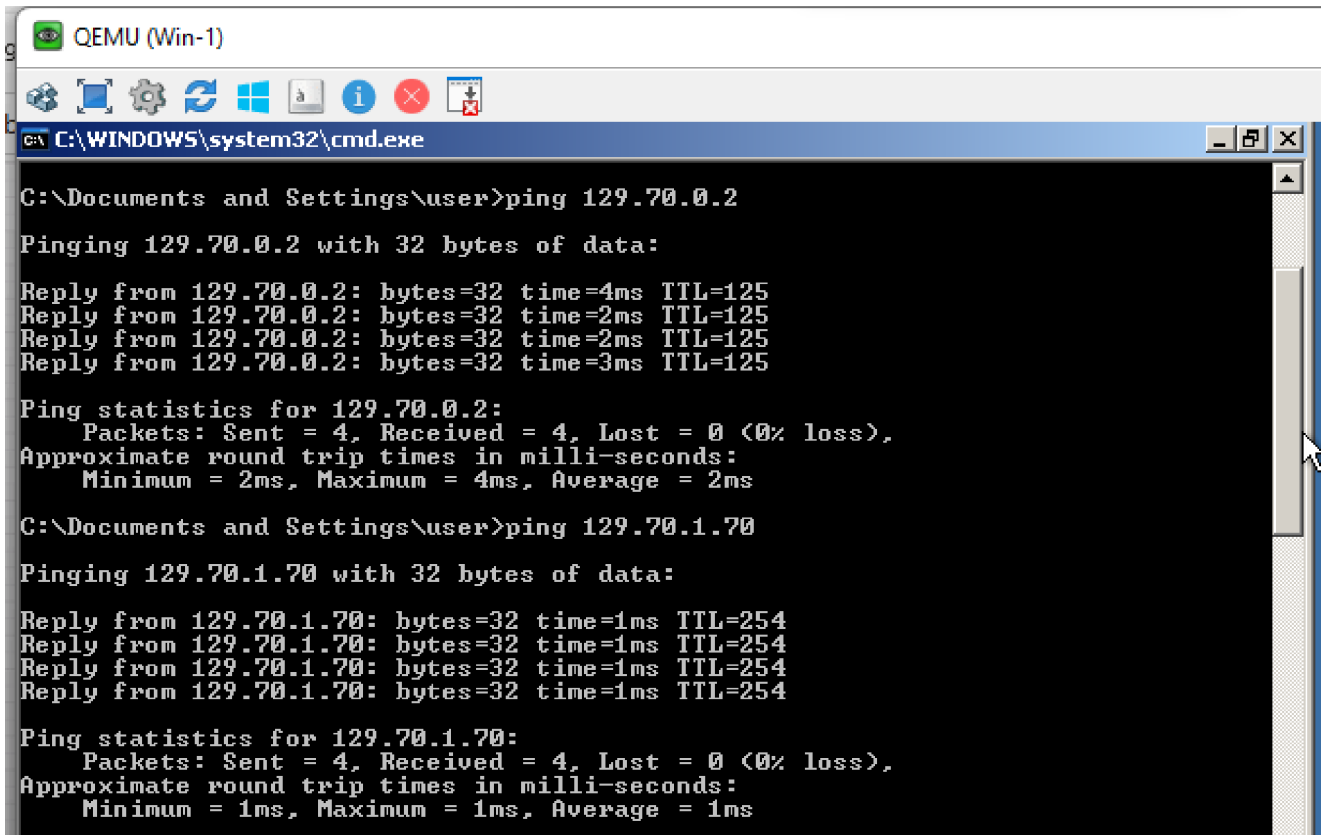


```
QEMU (Win-4)
C:\WINDOWS\system32\cmd.exe
Approximate round trip times in milli-seconds:
  Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>ping 129.70.1.69
Pinging 129.70.1.69 with 32 bytes of data:
Reply from 129.70.1.69: bytes=32 time=2ms TTL=253
Reply from 129.70.1.69: bytes=32 time=2ms TTL=253
Reply from 129.70.1.69: bytes=32 time=1ms TTL=253
Reply from 129.70.1.69: bytes=32 time=1ms TTL=253
Ping statistics for 129.70.1.69:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 1ms, Maximum = 2ms, Average = 1ms
C:\Documents and Settings\user>
C:\Documents and Settings\user>ping 12.0.1.144
Pinging 12.0.1.144 with 32 bytes of data:
Reply from 129.70.1.66: bytes=32 time=1ms TTL=254
Reply from 129.70.1.66: bytes=32 time=1ms TTL=254
Reply from 129.70.1.66: bytes=32 time=1ms TTL=254
Reply from 129.70.1.66: bytes=32 time=1ms TTL=254
Ping statistics for 12.0.1.144:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\Documents and Settings\user>ping 12.0.1.153
Pinging 12.0.1.153 with 32 bytes of data:
Reply from 12.0.1.153: bytes=32 time=1ms TTL=253
Reply from 12.0.1.153: bytes=32 time=1ms TTL=253
Reply from 12.0.1.153: bytes=32 time=1ms TTL=253
Reply from 12.0.1.153: bytes=32 time=1ms TTL=253
Ping statistics for 12.0.1.153:
  Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 1ms, Maximum = 1ms, Average = 1ms
C:\Documents and Settings\user>_
```

Figure 8 — Original project screenshot.

6. Technical Evidence — Screenshot 9

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



```
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ping 129.70.0.2

Pinging 129.70.0.2 with 32 bytes of data:

Reply from 129.70.0.2: bytes=32 time=4ms TTL=125
Reply from 129.70.0.2: bytes=32 time=2ms TTL=125
Reply from 129.70.0.2: bytes=32 time=2ms TTL=125
Reply from 129.70.0.2: bytes=32 time=3ms TTL=125

Ping statistics for 129.70.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>ping 129.70.1.70

Pinging 129.70.1.70 with 32 bytes of data:

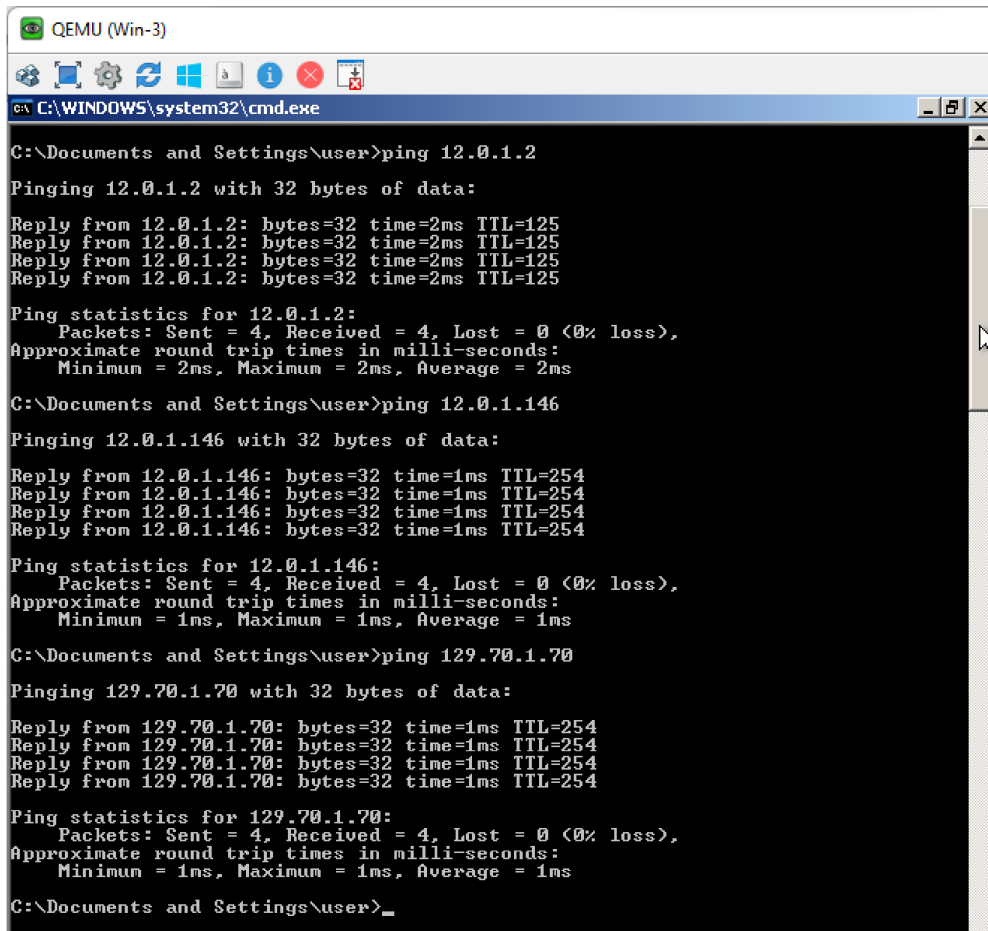
Reply from 129.70.1.70: bytes=32 time=1ms TTL=254
Reply from 129.70.1.70: bytes=32 time=1ms TTL=254
Reply from 129.70.1.70: bytes=32 time=1ms TTL=254
Reply from 129.70.1.70: bytes=32 time=1ms TTL=254

Ping statistics for 129.70.1.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms
```

Figure 9 — Original project screenshot.

6. Technical Evidence — Screenshot 10

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



```
C:\WINDOWS\system32\cmd.exe

C:\Documents and Settings\user>ping 12.0.1.2

Pinging 12.0.1.2 with 32 bytes of data:

Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125
Reply from 12.0.1.2: bytes=32 time=2ms TTL=125

Ping statistics for 12.0.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>ping 12.0.1.146

Pinging 12.0.1.146 with 32 bytes of data:

Reply from 12.0.1.146: bytes=32 time=1ms TTL=254
Reply from 12.0.1.146: bytes=32 time=1ms TTL=254
Reply from 12.0.1.146: bytes=32 time=1ms TTL=254
Reply from 12.0.1.146: bytes=32 time=1ms TTL=254

Ping statistics for 12.0.1.146:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Documents and Settings\user>ping 129.70.1.70

Pinging 129.70.1.70 with 32 bytes of data:

Reply from 129.70.1.70: bytes=32 time=1ms TTL=254
Reply from 129.70.1.70: bytes=32 time=1ms TTL=254
Reply from 129.70.1.70: bytes=32 time=1ms TTL=254
Reply from 129.70.1.70: bytes=32 time=1ms TTL=254

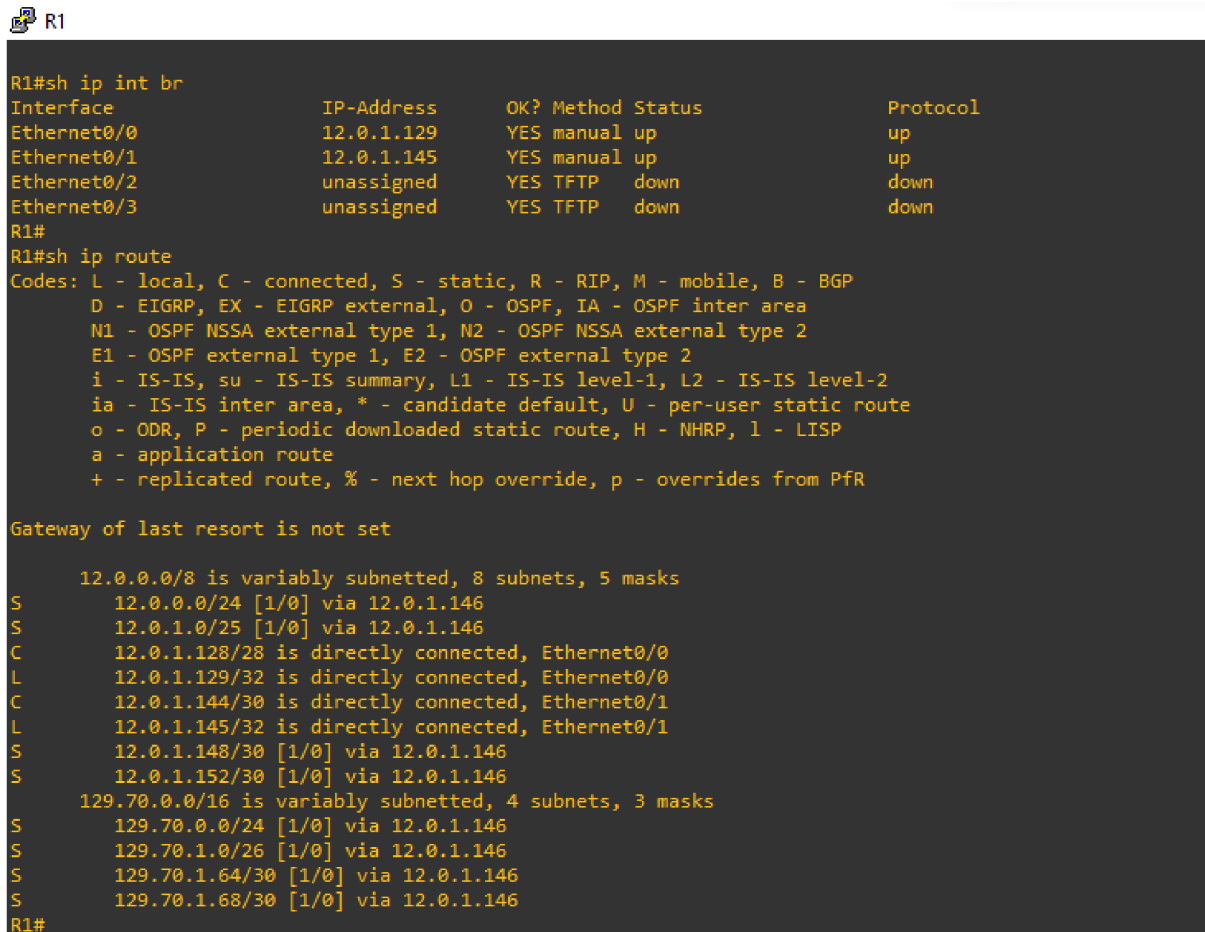
Ping statistics for 129.70.1.70:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 1ms, Average = 1ms

C:\Documents and Settings\user>_
```

Figure 10 — Original project screenshot.

6. Technical Evidence — Screenshot 11

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



```
R1#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              12.0.1.129      YES manual up          up
Ethernet0/1              12.0.1.145      YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
R1#
R1#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

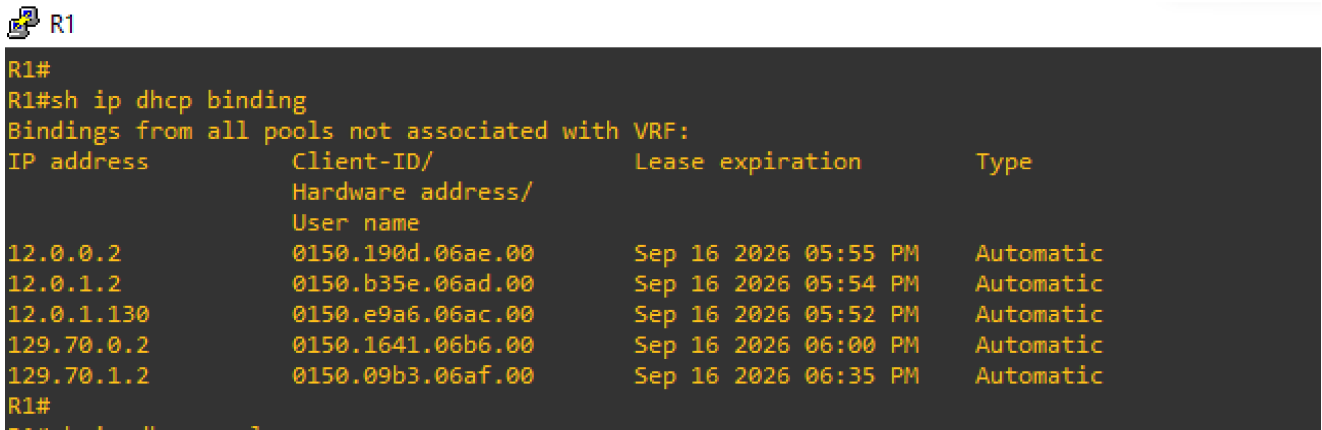
Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
S       12.0.0.0/24 [1/0] via 12.0.1.146
S       12.0.1.0/25 [1/0] via 12.0.1.146
C       12.0.1.128/28 is directly connected, Ethernet0/0
L       12.0.1.129/32 is directly connected, Ethernet0/0
C       12.0.1.144/30 is directly connected, Ethernet0/1
L       12.0.1.145/32 is directly connected, Ethernet0/1
S       12.0.1.148/30 [1/0] via 12.0.1.146
S       12.0.1.152/30 [1/0] via 12.0.1.146
    129.70.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       129.70.0.0/24 [1/0] via 12.0.1.146
S       129.70.1.0/26 [1/0] via 12.0.1.146
S       129.70.1.64/30 [1/0] via 12.0.1.146
S       129.70.1.68/30 [1/0] via 12.0.1.146
R1#
```

Figure 11 — Original project screenshot.

6. Technical Evidence — Screenshot 12

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



The screenshot shows a terminal window with a dark background and yellow text. The prompt is 'R1#'. The command entered is 'sh ip dhcp binding'. The output is a table of DHCP bindings. The first line of output is 'Bindings from all pools not associated with VRF:'. The table has four columns: 'IP address', 'Client-ID/ Hardware address/ User name', 'Lease expiration', and 'Type'. There are five rows of data. The first row is '12.0.0.2', '0150.190d.06ae.00', 'Sep 16 2026 05:55 PM', and 'Automatic'. The second row is '12.0.1.2', '0150.b35e.06ad.00', 'Sep 16 2026 05:54 PM', and 'Automatic'. The third row is '12.0.1.130', '0150.e9a6.06ac.00', 'Sep 16 2026 05:52 PM', and 'Automatic'. The fourth row is '129.70.0.2', '0150.1641.06b6.00', 'Sep 16 2026 06:00 PM', and 'Automatic'. The fifth row is '129.70.1.2', '0150.09b3.06af.00', 'Sep 16 2026 06:35 PM', and 'Automatic'. The prompt 'R1#' is visible at the bottom of the terminal window.

```
R1#
R1#sh ip dhcp binding
Bindings from all pools not associated with VRF:
IP address      Client-ID/      Lease expiration      Type
                Hardware address/
                User name
12.0.0.2        0150.190d.06ae.00  Sep 16 2026 05:55 PM  Automatic
12.0.1.2        0150.b35e.06ad.00  Sep 16 2026 05:54 PM  Automatic
12.0.1.130     0150.e9a6.06ac.00  Sep 16 2026 05:52 PM  Automatic
129.70.0.2     0150.1641.06b6.00  Sep 16 2026 06:00 PM  Automatic
129.70.1.2     0150.09b3.06af.00  Sep 16 2026 06:35 PM  Automatic
R1#
```

Figure 12 — Original project screenshot.

6. Technical Evidence — Screenshot 13

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.

 R1

```
R1#sh ip dhcp pool

Pool PC1 :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 14
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  12.0.1.131         12.0.1.129 - 12.0.1.142    1

Pool PC2 :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 126
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  12.0.1.3           12.0.1.1 - 12.0.1.126    1

Pool PC3 :
  Utilization mark (high/low)    : 100 / 0
  Subnet size (first/next)       : 0 / 0
  Total addresses                 : 254
  Leased addresses                : 1
  Pending event                  : none
  1 subnet is currently in the pool :
  Current index      IP address range      Leased addresses
  12.0.0.3           12.0.0.1 - 12.0.0.254    1
```

Figure 13 — Original project screenshot.

6. Technical Evidence — Screenshot 14

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.

```
R1
Pending event      : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
12.0.1.131         12.0.1.129 - 12.0.1.142         1

Pool PC2 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses             : 126
Leased addresses            : 1
Pending event               : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
12.0.1.3           12.0.1.1 - 12.0.1.126    1

Pool PC3 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses             : 254
Leased addresses            : 1
Pending event               : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
12.0.0.3           12.0.0.1 - 12.0.0.254    1

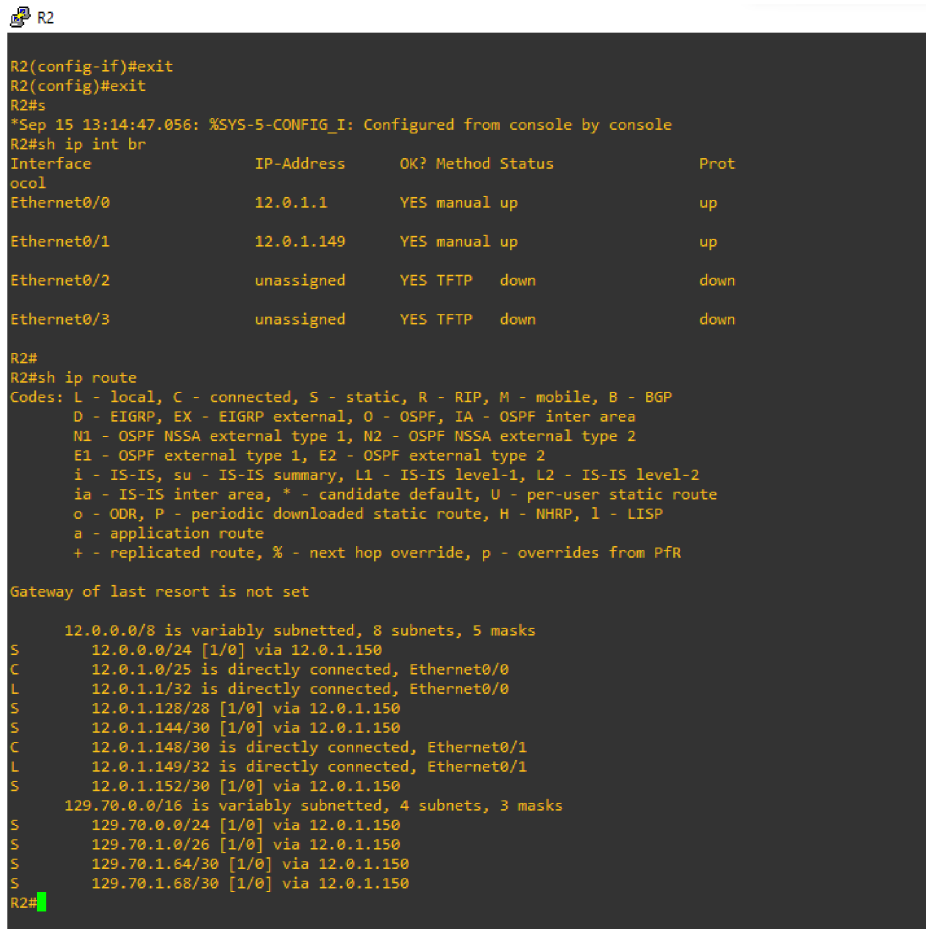
Pool PC4 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses             : 62
Leased addresses            : 1
Pending event               : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
129.70.1.3         129.70.1.1 - 129.70.1.62    1

Pool PC5 :
Utilization mark (high/low) : 100 / 0
Subnet size (first/next)    : 0 / 0
Total addresses             : 254
Leased addresses            : 1
Pending event               : none
1 subnet is currently in the pool :
Current index      IP address range      Leased addresses
129.70.0.3         129.70.0.1 - 129.70.0.254    1
R1#
R1#
```

Figure 14 — Original project screenshot.

6. Technical Evidence — Screenshot 15

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



```
R2
R2(config-if)#exit
R2(config)#exit
R2#
*Sep 15 13:14:47.056: %SYS-5-CONFIG_I: Configured from console by console
R2#sh ip int br
Interface                                IP-Address      OK? Method Status        Prot
-----                                -
Ethernet0/0                             12.0.1.1        YES manual up            up
Ethernet0/1                             12.0.1.149     YES manual up            up
Ethernet0/2                             unassigned      YES TFTP  down          down
Ethernet0/3                             unassigned      YES TFTP  down          down

R2#
R2#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

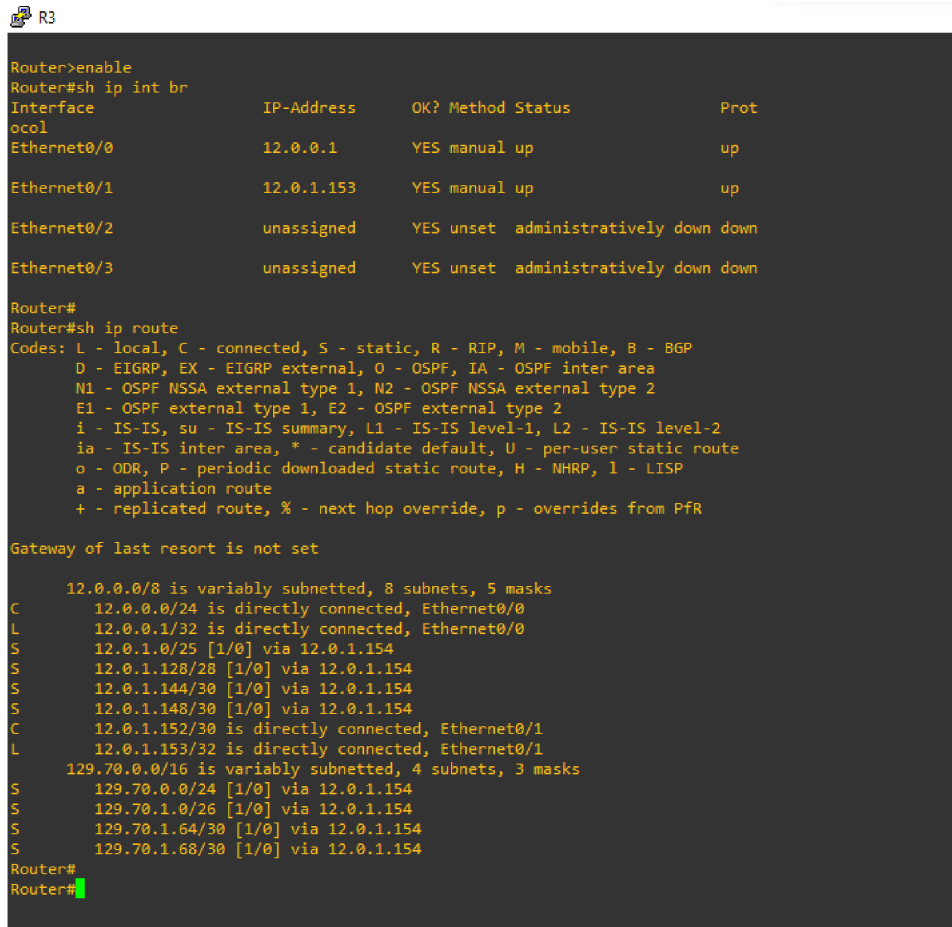
Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
S       12.0.0.0/24 [1/0] via 12.0.1.150
C       12.0.1.0/25 is directly connected, Ethernet0/0
L       12.0.1.1/32 is directly connected, Ethernet0/0
S       12.0.1.128/28 [1/0] via 12.0.1.150
S       12.0.1.144/30 [1/0] via 12.0.1.150
C       12.0.1.148/30 is directly connected, Ethernet0/1
L       12.0.1.149/32 is directly connected, Ethernet0/1
S       12.0.1.152/30 [1/0] via 12.0.1.150
    129.70.0.0/16 is variably subnetted, 4 subnets, 3 masks
S       129.70.0.0/24 [1/0] via 12.0.1.150
S       129.70.1.0/26 [1/0] via 12.0.1.150
S       129.70.1.64/30 [1/0] via 12.0.1.150
S       129.70.1.68/30 [1/0] via 12.0.1.150
R2#
```

Figure 15 — Original project screenshot.

6. Technical Evidence — Screenshot 16

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



```
R3
Router>enable
Router#sh ip int br
Interface IP-Address OK? Method Status Prot
ocol
Ethernet0/0 12.0.0.1 YES manual up up
Ethernet0/1 12.0.1.153 YES manual up up
Ethernet0/2 unassigned YES unset administratively down down
Ethernet0/3 unassigned YES unset administratively down down

Router#
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PFR

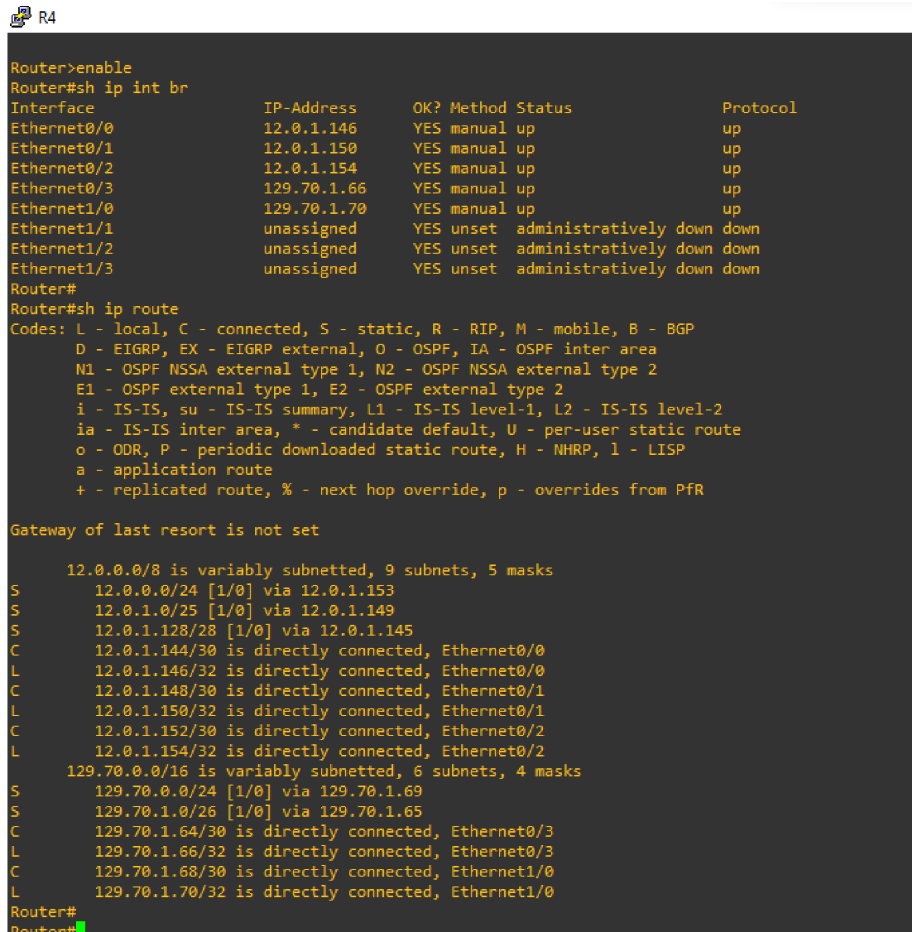
Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 8 subnets, 5 masks
C 12.0.0.0/24 is directly connected, Ethernet0/0
L 12.0.0.1/32 is directly connected, Ethernet0/0
S 12.0.1.0/25 [1/0] via 12.0.1.154
S 12.0.1.128/28 [1/0] via 12.0.1.154
S 12.0.1.144/30 [1/0] via 12.0.1.154
S 12.0.1.148/30 [1/0] via 12.0.1.154
C 12.0.1.152/30 is directly connected, Ethernet0/1
L 12.0.1.153/32 is directly connected, Ethernet0/1
129.70.0.0/16 is variably subnetted, 4 subnets, 3 masks
S 129.70.0.0/24 [1/0] via 12.0.1.154
S 129.70.1.0/26 [1/0] via 12.0.1.154
S 129.70.1.64/30 [1/0] via 12.0.1.154
S 129.70.1.68/30 [1/0] via 12.0.1.154
Router#
Router#
```

Figure 16 — Original project screenshot.

6. Technical Evidence — Screenshot 17

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



```
R4
Router>enable
Router#sh ip int br
Interface IP-Address OK? Method Status Protocol
Ethernet0/0 12.0.1.146 YES manual up up
Ethernet0/1 12.0.1.150 YES manual up up
Ethernet0/2 12.0.1.154 YES manual up up
Ethernet0/3 129.70.1.66 YES manual up up
Ethernet1/0 129.70.1.70 YES manual up up
Ethernet1/1 unassigned YES unset administratively down down
Ethernet1/2 unassigned YES unset administratively down down
Ethernet1/3 unassigned YES unset administratively down down
Router#
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PFR

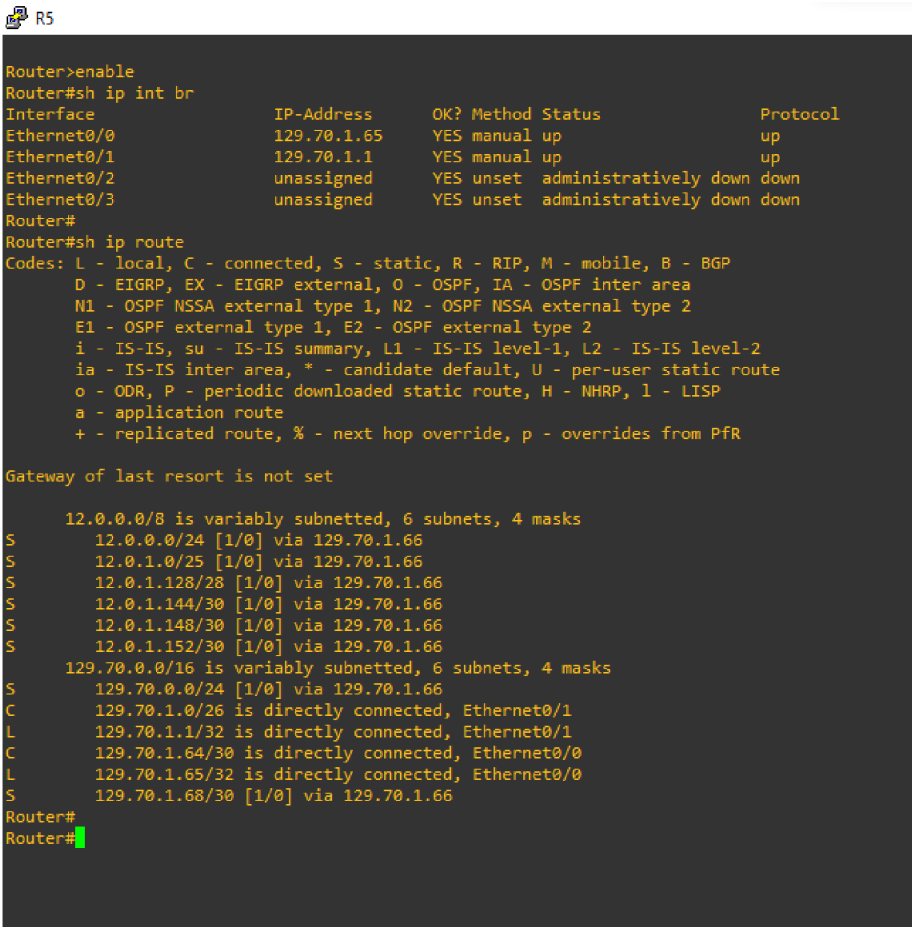
Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 9 subnets, 5 masks
S       12.0.0.0/24 [1/0] via 12.0.1.153
S       12.0.1.0/25 [1/0] via 12.0.1.149
S       12.0.1.128/28 [1/0] via 12.0.1.145
C       12.0.1.144/30 is directly connected, Ethernet0/0
L       12.0.1.146/32 is directly connected, Ethernet0/0
C       12.0.1.148/30 is directly connected, Ethernet0/1
L       12.0.1.150/32 is directly connected, Ethernet0/1
C       12.0.1.152/30 is directly connected, Ethernet0/2
L       12.0.1.154/32 is directly connected, Ethernet0/2
    129.70.0.0/16 is variably subnetted, 6 subnets, 4 masks
S       129.70.0.0/24 [1/0] via 129.70.1.69
S       129.70.1.0/26 [1/0] via 129.70.1.65
C       129.70.1.64/30 is directly connected, Ethernet0/3
L       129.70.1.66/32 is directly connected, Ethernet0/3
C       129.70.1.68/30 is directly connected, Ethernet1/0
L       129.70.1.70/32 is directly connected, Ethernet1/0
Router#
```

Figure 17 — Original project screenshot.

6. Technical Evidence — Screenshot 18

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



```
Router>enable
Router#sh ip int br
Interface                IP-Address      OK? Method Status        Protocol
Ethernet0/0              129.70.1.65     YES manual up            up
Ethernet0/1              129.70.1.1      YES manual up            up
Ethernet0/2              unassigned      YES unset  administratively down down
Ethernet0/3              unassigned      YES unset  administratively down down
Router#
Router#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

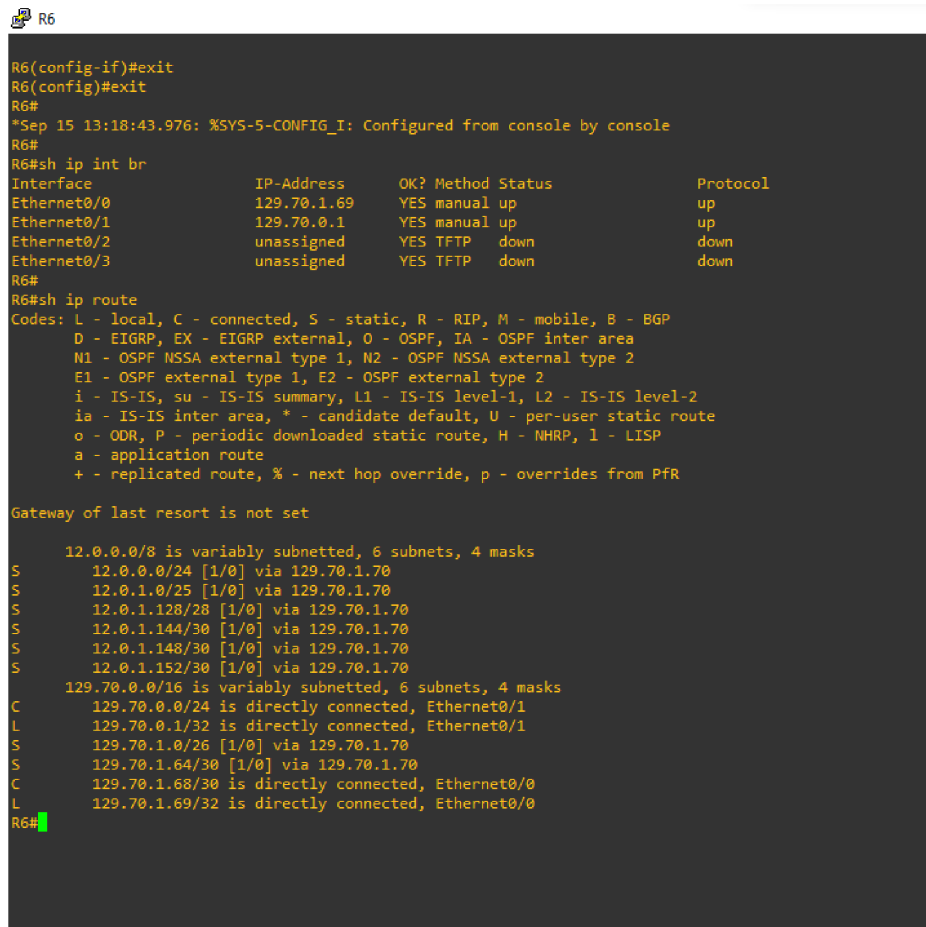
Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       12.0.0.0/24 [1/0] via 129.70.1.66
S       12.0.1.0/25 [1/0] via 129.70.1.66
S       12.0.1.128/28 [1/0] via 129.70.1.66
S       12.0.1.144/30 [1/0] via 129.70.1.66
S       12.0.1.148/30 [1/0] via 129.70.1.66
S       12.0.1.152/30 [1/0] via 129.70.1.66
    129.70.0.0/16 is variably subnetted, 6 subnets, 4 masks
S       129.70.0.0/24 [1/0] via 129.70.1.66
C       129.70.1.0/26 is directly connected, Ethernet0/1
L       129.70.1.1/32 is directly connected, Ethernet0/1
C       129.70.1.64/30 is directly connected, Ethernet0/0
L       129.70.1.65/32 is directly connected, Ethernet0/0
S       129.70.1.68/30 [1/0] via 129.70.1.66
Router#
Router#
```

Figure 18 — Original project screenshot.

6. Technical Evidence — Screenshot 19

Original project evidence reproduced at high resolution. The screenshot is retained as submitted so command output, addressing information and network-state details can be inspected directly.



```
R6
R6(config-if)#exit
R6(config)#exit
R6#
*Sep 15 13:18:43.976: %SYS-5-CONFIG_I: Configured from console by console
R6#
R6#sh ip int br
Interface                IP-Address      OK? Method Status      Protocol
Ethernet0/0              129.70.1.69     YES manual up          up
Ethernet0/1              129.70.0.1      YES manual up          up
Ethernet0/2              unassigned      YES TFTP  down        down
Ethernet0/3              unassigned      YES TFTP  down        down
R6#
R6#sh ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PFR

Gateway of last resort is not set

    12.0.0.0/8 is variably subnetted, 6 subnets, 4 masks
S       12.0.0.0/24 [1/0] via 129.70.1.70
S       12.0.1.0/25 [1/0] via 129.70.1.70
S       12.0.1.128/28 [1/0] via 129.70.1.70
S       12.0.1.144/30 [1/0] via 129.70.1.70
S       12.0.1.148/30 [1/0] via 129.70.1.70
S       12.0.1.152/30 [1/0] via 129.70.1.70
    129.70.0.0/16 is variably subnetted, 6 subnets, 4 masks
C       129.70.0.0/24 is directly connected, Ethernet0/1
L       129.70.0.1/32 is directly connected, Ethernet0/1
S       129.70.1.0/26 [1/0] via 129.70.1.70
S       129.70.1.64/30 [1/0] via 129.70.1.70
C       129.70.1.68/30 is directly connected, Ethernet0/0
L       129.70.1.69/32 is directly connected, Ethernet0/0
R6#
```

Figure 19 — Original project screenshot.

7. Conclusion

This project demonstrates a practical CCNA-level workflow for building and validating an IPv4 network. The work covers subnetting and addressing, DHCP-based host configuration, router verification, routing and connectivity testing.

Key learning outcomes

- Understanding how subnetting separates a larger address space into usable networks.
- Understanding how DHCP simplifies client IP configuration.
- Understanding how routers connect different IPv4 networks.
- Using Cisco IOS show commands as evidence during troubleshooting.
- Using ipconfig and ping to verify the final state from an end-device perspective.

Resume-ready project statement

CCNA Networking Lab — IPv4 Subnetting & DHCP: Designed and validated a routed IPv4 network using subnetting and DHCP; verified host addressing, router interfaces and routing information; and performed end-to-end connectivity testing with Cisco IOS and Windows CLI tools.

This report follows the organization and presentation style of the attached CCNA project example while using the technical evidence from the supplied Subnetting with DHCP project.